

Fabien PETITJEAN

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<https://tolokoban.github.io/portfolio/>

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Web developer fullstack

Visualization engineer

Education

1997: Master degree in computer Science at INSA [[link](#)]

2010: Bachelor in Mathematics at Besançon University [[link](#)]

Employment *(see details on the next page)*

January 2025.....now: Blue Brain Project (EPFL)
July 2019.....Dec. 2024: Blue Brain Project (EPFL)
2006.....now: TrailPassion (non-profit organisation)

June 2006.....June 2019: Lombard Odier Private Bank
Mai 2003.....May 2006: SOLOG (Switzerland)
July 2002.....April 2003: Reuters
July 2001.....June 2002: SOLOG (Switzerland)
Sept 2000.....June 2001: SOLOG (France)
Dec 1999.....Aug 2000: ATOS Origin
July 1996.....Nov 1999: INSERM

Skills

Human languages: French (father tongue), Italian (mother tongue), English (7 years in an english environment)

Computer languages: Typescript, WebGL2, SVG, MathML, PHP, NodeJS, Python3, Rust, C#, C++, JAVA, LaTeX

Tools: React, Rspack, Webpack, Blender, Krita, VS Code, Tauri, Electron, git, threeJS

Hobbies

Sport: Trail-running (GR20 2026, GR5 2024, Diagonale des fous 2015, UTMB 2005, ...), Basket-Ball, free-falling (72 jumps)

Media: 3D modeling, VFX and video editing with Blender, digital painting with Krita, short film maker (Le Morbide)

Games for children: [Ansy-2025](#)

Readings: Metal Hurlant, Martin Gardener, Sam Loyd (I love puzzles), Raymon Smullyan, Donald Knuth, Russel/Norvig, Liu Xin (three body problem), Lovecraft, ...

More details on the next page...

Employment details

January 2026 – now: Open Brain Institute ([OBI](#))

When the Blue Brain Project ended, 35 people over 150 were selected to continue the adventure in a startup: Open Brain Institute.

- **Core Web App** ([link](#)): a cloud based and AI assisted application to help neuro-scientists o experiment on the brain of flies, mice, rats and (soon) humans. I work in a team of 4 fullstack developers. [TYPESCRIPT](#) [REACT](#) [WEBGL](#) [WEBPACK](#) [DOCKER](#)
- **MorphoViewer** ([link](#)): the library used in Core Web App to deal with all the 3D visualizations. I am the sole developer of this library. [TYPESCRIPT](#) [REACT](#) [TAURI](#) [RUST](#) [WEBGL](#) [RSPACK](#)

July 2019 – Dec. 2025: Blue Brain Project ([EPFL](#))

The Blue Brain Project aimed to simulate a full mouse brain (80 million neurons). I worked in the Visualization Team to provide tools that help scientists to validate their models visually.

- **Circuit Studio** ([link](#)): Virtual microscope into the simulated mouse's brain. [TYPESCRIPT](#) [REACT](#) [WEBGL](#) [WEBPACK](#) [DOCKER](#)
- **Open Brain Platform** ([link](#)): the app that summarizes 18 years of the Blue Brain Project. [TYPESCRIPT](#) [REACT](#) [WEBGL](#) [NEXTJS](#)
- **Hippocampus coordinates** ([link](#)): Visual validation of a curved coordinates system. [TYPESCRIPT](#) [REACT](#) [WEBGL](#) [NEXTJS](#)

2006 – now: trail-passion.net

non-profit association with three members dedicated in outdoor software development.

- **Circuits** ([link](#)): Mobile application (iOS & Android) sold to tourist office. The client can update content via a web interface and the mobile app is updated automatically. [PHP](#) [MYSQL](#) [LEAFLET](#) [TYPESCRIPT](#) [REACT](#)
- **TrailPassion website** ([link](#)): display and edit geographical information on different maps. [OPENSTREETMAP](#) [WEBGL](#) [PYTHON](#) [NODEJS](#) [DOCKER](#) [SRTM](#)
- **Race timer** ([link](#)): Cheap system to time mountain races. You just type the bib race of runners passing checkpoints and the ranking is updated on the website.

June 2006 – June 2019: Lombard Odier Private Bank

- **NegoceIDL**: [JAVA](#) program parsing IDL files provided by CORBA servers, and making [VB6](#), [C#](#) and [C++](#) code for clients.
- **AntiRegressor**: Regression test software for [CORBA](#) servers. It provides a “visual programming” interface and can be used by testers with no coding skills.
- **Generic Reports**: based on a rich XML language an on [XSLT](#) templates , it produces HTML, Excel, LibreOffice, XML and PDF files for financial reports. We used a [LATEX](#) standalone installation to build PDF quickly (TexLive).
- **SalesOrders & VNegoce**: two orders management systems written in [VB6](#), [C#](#) and [C++](#).
- **Fair allocation**: algorithm of fair trades distribution. Prototyped in [PYTHON](#), implemented in C++ and VB6

Mai 2003 – May 2006: SOLOG (Switzerland)

- **DB migration:** BNP Paribas gave me one week to rewrite more than 500 SQL queries because of a deep change in the DB schema. I wrote a lexer/scanner to do this automatically. `SQL` `PERL`.
- **Market Orders Packaging:** pack compatible orders in a single one to pay less fees. `MERCATOR`, SQL.

July 2002 – April 2003: Reuters

- **PermLib:** Asynchronous C++ library delivering quotes and permissions on RIC instruments. Tibco RendezVous, RogueWave, STL.
- **QuoteAnalyser:** software to test mathematical models for quote variations previsions. Python 2, numpy.

July 2001 – June 2002: SOLOG (Switzerland)

- **ITC/ONU** [<http://jitap.org>], Euronews and SITA in Geneva to develop web sites. HTML, CSS, Javascript, `PHP`.

Sept 2000 – June 2001: SOLOG (France)

- **SOLOG website:** coding and design. HTML, CSS, Javascript, `PHP`.

Dec 1999 – Aug 2000: ATOS Origin

- **E-Deploy:** (mandated to Hewlett Packard) Computers purchase web site dedicated to big customers (British Telecom, SNCF, ...). HTML, CSS, Javascript.

July 1996 – Nov 1999: INSERM

- **MATIS** (Mathematical Tools Integrating System): Graphical language where the user connects nodes and the system executes them on available Matlab servers on the local network. Matlab, Visual Basic, ASP.